

HOMEWORK 2  
INTRODUCTION TO COMPLEX ANALYSIS  
(MATH 4230-40), SPRING 2022

DUE: 02/04/2022

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SECTION 1.3 - SOME ELEMENTARY FUNCTIONS

1. Find all the values of:
    - (a)  $(-i)^i$
    - (b)  $(1+i)^{1+i}$
  2. Is it true that  $|a^b| = |a|^{|b|}$  for all  $a, b \in \mathbb{C}$ ? Why or why not?
  3. Show that under the map  $z \mapsto z^2$ , lines parallel to the real axis are mapped to parabolas.
  4. Show that under a branch of the map  $z \mapsto \sqrt{z}$ , lines parallel to the real axis are mapped to hyperbolas.
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SECTION 1.4 - CONTINUOUS FUNCTIONS

5. Prove that the complement of any finite set of points in  $\mathbb{C}$  is an open set.
6. For what  $z \in \mathbb{C}$  does the sequence  $(z_n)$  given by  $z_n = nz^n$  converge?
7. For each of the following sets, state whether or not it is connected and whether or not it is compact:
  - (a)  $\{z \in \mathbb{C} : 1 \leq |z| \leq 2\}$
  - (b)  $\{z \in \mathbb{C} : |z| \leq 3, |\operatorname{Re} z| \geq 1\}$
  - (c)  $\{z \in \mathbb{C} : |\operatorname{Re} z| \leq 1\}$
8. Use the fact that a function is continuous if and only if the preimage of every open set is open to show that a composition of two continuous functions is continuous.